

AIPP LAB 15

Name: **MOHAMMED NUMAAN AHMED**

Hall Ticket No.: **2503B05127**

Date: **25/11/2025 (Week 5)**

Completed this Assignment using VS CODE AI Integration.

Task 1:

Ask Use Python (or Node.js/JavaScript) to connect to a public API (e.g., OpenWeatherMap or JSONPlaceholder). Send a simple GET request to retrieve data. Display the response in a readable format (pretty JSON). Prompt: write a python function that display weather details of a city using weather api without error handling. Display weather details as JSON output.

CODE:

```
TASK15.1.py 1
C: > Users > ACER > OneDrive > Documents > MAHVISH M.TECH AIPP LABS > ASSIGNMENT 15 >
1
2 import requests
3 import json
4 from datetime import datetime
5 API_KEY = "e6c6063509fc4d450cde0ca4414b3a9f"
6 BASE_URL = "https://api.openweathermap.org/data/2.5/weather"
7 def get_weather(city):
8     """
9     Fetch weather details for a city using OpenWeather API.
10    Includes error handling, JSON display and file storage.
11    """
12    params = {
13        "q": city,
14        "appid": API_KEY,
15        "units": "metric"
16    }
17    try:
18        response = requests.get(BASE_URL, params=params, timeout=5)
19        # Raise error if HTTP response is not OK
20        response.raise_for_status()
21        data = response.json()
22        # Extract fields
23        temperature = data["main"]["temp"]
24        humidity = data["main"]["humidity"]
25        weather_desc = data["weather"][0]["description"].title()
26        # User-friendly output for Task 3 & 4
27        print("\n----- WEATHER REPORT -----")
28        print(f"City: {city.title()}")
29        print(f"Temperature: {temperature}°C")
30        print(f"Humidity: {humidity}%")
31        print(f"Weather: {weather_desc}")
32        # Append results to results.txt (Task 5)
33        entry = {
34            "city": city.title(),
35            "temperature": temperature,
36            "humidity": humidity,
37            "weather": weather_desc,
```

```

TASK15.1.py 1
C:\Users\ACER> OneDrive\Documents\MAHVISH M.TECH AIPP LABS\ASSIGNMENT 15> TASK15.1.py
7 def get_weather(city):
37     "weather": weather_desc,
38     "time": str(datetime.now())
39 }
40 with open("results.txt", "a") as file:
41     file.write(json.dumps(entry) + "\n")
42     return entry # For test cases
43 except requests.exceptions.Timeout:
44     print("Error: Request timed out, check your network.")
45 except requests.exceptions.ConnectionError:
46     print("Error: Could not connect to API. Check internet connection.")
47 except requests.exceptions.HTTPError:
48     print("Error: Invalid city name or API key.")
49 except Exception as e:
50     print("Unexpected error:", str(e))
51 return None
52 # CALL THE FUNCTION HERE
53 get_weather("Hyderabad") # ← required!
54 # CALL THE FUNCTION HERE
55 get_weather("Delhi") # ← required!
56 # CALL THE FUNCTION HERE
57 get_weather("Bangalore") # ← required!

```

OUTPUT:

```

===== FULL API JSON RESPONSE =====
{
  "coord": {
    "lon": 78.4744,
    "lat": 17.3753
  },
  "weather": [
    {
      "id": 721,
      "main": "Haze",
      "description": "haze",
      "icon": "50n"
    }
  ],
  "base": "stations",
  "main": {
    "temp": 23.23,
    "feels_like": 23.38,
    "temp_min": 21.73,
    "temp_max": 23.23,
    "pressure": 1016,
    "humidity": 68,
    "sea_level": 1016,
    "grnd_level": 952
  },
  "visibility": 2500,
  "wind": {
    "speed": 2.06,
    "deg": 110
  },
  "clouds": {
    "all": 40
  },
  "dt": 1764083372,
  "sys": {
    "type": 1,
    "id": 9214,
    "country": "IN",
    "sunrise": 1764032203,
    "sunset": 1764072586
  },
  "timezone": 19800,
  "id": 1269843,
  "name": "Hyderabad",
  "cod": 200
}

===== WEATHER REPORT =====
City: Hyderabad
Temperature: 23.23°C
Humidity: 68%
Weather: Haze
PS C:\Users\ACER\OneDrive\Documents\MAHVISH M.TECH AIPP LABS\ASSIGNMENT 15>

```

Task 2:

Prompt: write a python function that display weather details of a city using weather api with error handling. Display weather details as JSON output.

CODE:

```

import requests # type: ignore
import json
API_KEY = "e6c6083509fc4d450cde0ca4414b3f"
def get_weather_with_errors(city):
    url = f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units=metric"
    try:
        response = requests.get(url, timeout=5)
        response.raise_for_status()
        data = response.json()

        print(json.dumps(data, indent=4))
        return data
    except requests.exceptions.Timeout:
        print("Error: API request timed out.")
    except requests.exceptions.ConnectionError:
        print("Error: Could not connect to API. Check your internet.")
    except requests.exceptions.HTTPError:
        print("Error: Invalid city or API key.")
    except Exception as e:
        print("Unexpected Error:", str(e))
    return None
get_weather_with_errors("Hyderabad")

```

OUTPUT:

```

PS C:\Users\ACER\OneDrive\Documents\MAHVISH M.TECH AIPP LABS\ASSIGNMENT 15> c:: cd 'c:\Users\ACER\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.python\python\python.exe' 'c:\Users\ACER\OneDrive\Documents\MAHVISH M.TECH AIPP LABS\ASSIGNMENT 15\TASK15.2.py'
Error: Invalid city or API key.
PS C:\Users\ACER\OneDrive\Documents\MAHVISH M.TECH AIPP LABS\ASSIGNMENT 15>

```

Task 3:

Extract and Display Specific Data

• Instructions:

1. From the API response (e.g., weather API), extract specific fields (temperature, humidity, description).
2. Display them in a user-friendly format (not raw JSON)

CODE:

```

> Users > ACER > OneDrive > Documents > MAHVISH M.TECH AI PP LABS > ASSIGNMENT 15 > TASK15.3.py > get_weather_with_errs
3 import json
4 API_KEY = "e6c6083509fc4d450cde0ca4414b3a9f"
5 def get_weather_with_errors(city):
6     url = f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units=metric"
7     try:
8         response = requests.get(url, timeout=5)
9         response.raise_for_status()
10        data = response.json()
11        print(json.dumps(data, indent=4))
12        return data
13    except requests.exceptions.Timeout:
14        print("Error: API request timed out.")
15    except requests.exceptions.ConnectionError:
16        print("Error: Could not connect to API. Check your internet.")
17    except requests.exceptions.HTTPError:
18        print("Error: Invalid city or API key.")
19    except Exception as e:
20        print("Unexpected Error:", str(e))
21    return None
22 get_weather_with_errors("Hyderabad")
23 def get_weather_pretty(city):
24     data = get_weather_with_errors(city)
25     if data is None:
26         return None
27     city_name = data["name"]
28     temp = data["main"]["temp"]
29     hum = data["main"]["humidity"]
30     desc = data["weather"][0]["description"].title()
31     print(f"City: {city_name}")
32     print(f"Temperature: {temp}°C")
33     print(f"Humidity: {hum}%")
34     print(f"Weather: {desc}")
35     return {"city": city_name, "temp": temp, "humidity": hum, "weather": desc}
36 res = get_weather_pretty("London")
37 assert "city" in res
38 assert "temp" in res
39 assert "weather" in res

```

OUTPUT:

```

PS C:\Users\ACER\OneDrive\Documents\MAHVISH M.TECH AI PP LABS\ASSIGNMENT 15\TASK15.3.py> python3.12.exe 'c:\Users\ACER\OneDrive\Documents\MAHVISH M.TECH AI PP LABS\ASSIGNMENT 15\TASK15.3.py'
{
  "coord": {
    "lon": 78.4744,
    "lat": 17.3753
  },
  "weather": [
    {
      "id": 721,
      "main": "haze",
      "description": "haze",
      "icon": "50a"
    }
  ],
  "base": "stations",
  "main": {
    "temp": 21.23,
    "feels_like": 21.45,
    "temp_min": 21.23,
    "temp_max": 21.73,
    "pressure": 1016,
    "humidity": 78,
    "sea_level": 1016,
    "grnd_level": 952
  },
  "visibility": 3000,
  "wind": {
    "speed": 2.06,
    "deg": 110
  },
  "clouds": {
    "all": 80
  },
  "dt": 1764084370,
  "sys": {
    "type": 1,
    "id": 9214,
    "country": "IN",
    "sunrise": 1764032203,
    "sunset": 1764072586
  },
  "timezone": 19800,
  "id": 1269843,
  "name": "Hyderabad",
  "cod": 200
}
{
  "coord": {
    "lon": -0.1257,
    "lat": 51.5085
  },
  "weather": [
    {
      "id": 803,
      "main": "Clouds",
      "description": "broken clouds",
      "icon": "04d"
    }
  ],
  "base": "stations",
  "main": {
    "temp": 6.58,
    "feels_like": 4.02,
    "temp_min": 5.95,
    "temp_max": 7.65,
    "pressure": 1015,
    "humidity": 81,
    "sea_level": 1015,
    "grnd_level": 1011
  },
  "visibility": 10000,
  "wind": {
    "speed": 3.6,

```

```

        "deg": 300
    },
    "clouds": {
        "all": 75
    },
    "dt": 1764084132,
    "sys": {
        "type": 2,
        "id": 2075535,
        "country": "GB",
        "sunrise": 1764056116,
        "sunset": 1764086409
    },
    "timezone": 0,
    "id": 2643743,
    "name": "London",
    "cod": 200
}
City: London
Temperature: 6.58°C
Humidity: 81%
Weather: Broken Clouds
C:\PS C:\Users\ACER\OneDrive\Documents\MAHVISH M.TECH AIPP LABS\ASSIGNMENT 15>

```

Task 4:

Build a Function with a Parameter. Write a python function that displays weather details of a city using weather api with error handling. Display weather details in user friendly forma

CODE:

```

> Users > ACER > OneDrive > Documents > MAHVISH M.TECH AIPP LABS > ASSIGNMENT 15 > TASK15.4.py > fetch_weather
1  import requests # type: ignore
2  import json
3  API_KEY = "e6c6083509fc4d45ecde8ca4414b3a9f"
4  def get_weather_with_errors(city):
5      url = f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units=metric"
6      try:
7          response = requests.get(url, timeout=5)
8          response.raise_for_status()
9          data = response.json()
10         print(json.dumps(data, indent=4))
11         return data
12     except requests.exceptions.Timeout:
13         print("Error: API request timed out.")
14     except requests.exceptions.ConnectionError:
15         print("Error: Could not connect to API. Check your internet.")
16     except requests.exceptions.HTTPError:
17         print("Error: Invalid city or API key.")
18     except Exception as e:
19         print("Unexpected Error:", str(e))
20     return None
21 def fetch_weather(city):
22     data = get_weather_with_errors(city)
23     if data is None:
24         print("Error: City not found. Please enter a valid city.")
25         return None
26     result = {
27         "city": data["name"],
28         "temp": data["main"]["temp"],
29         "humidity": data["main"]["humidity"],
30         "weather": data["weather"][0]["description"].title()
31     }
32     print(result)
33     return result
34 assert fetch_weather("New York")["city"] == "New York"
35 assert fetch_weather("xyz123") is None
36 assert isinstance(fetch_weather("Delhi"), dict)
37

```

OUTPUT:

```
{
  "coord": {
    "lon": 77.2167,
    "lat": 28.6667
  },
  "weather": [
    {
      "id": 721,
      "main": "Haze",
      "description": "haze",
      "icon": "500"
    }
  ],
  "base": "stations",
  "main": {
    "temp": 17.05,
    "feels_like": 16.35,
    "temp_min": 17.05,
    "temp_max": 17.05,
    "pressure": 1017,
    "humidity": 59,
    "sea_level": 1017,
    "grnd_level": 991
  },
  "visibility": 2000,
  "wind": {
    "speed": 1.03,
    "deg": 240
  },
  "clouds": {
    "all": 0
  },
  "dt": 1764084575,
  "sys": {
    "type": 1,
    "id": 9165,
    "country": "IN",
    "sunrise": 1764033716,
    "sunset": 1764071677
  }
}
```

```
},
"timezone": 19800,
"id": 1273294,
"name": "Delhi",
"cod": 200
}
{"city": 'Delhi', 'temp': 17.05, 'humidity': 59, 'weather': 'Haze'}
© PS C:\Users\ACER\OneDrive\Documents\WVWVISH M.TECH AI&PP LABS\ASSIGNMENT 15>
```

Task 5:

Prompt: write a python function that display weather details of a city using weather api with error handling. Display weather details as JSON output. Store the weather details in current directory as text file, every run output will append.

CODE:

TASK15.1.py TASK15.2.py TASK15.3.py TASK15.4.py TASK15.5.py X

C: > Users > ACER > OneDrive > Documents > MAHVISH M.TECH AIPP LABS > ASSIGNMENT 15 > TASK15.5.py > save_weather

```

1 import requests # type: ignore
2 import json
3 API_KEY = "e6c683509fc4d450cde8ca4414b3a9f"
4 def get_weather_with_errors(city):
5     url = f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units=metric"
6     try:
7         response = requests.get(url, timeout=5)
8         response.raise_for_status()
9         data = response.json()
10        print(json.dumps(data, indent=4))
11        return data
12    except requests.exceptions.Timeout:
13        print("Error: API request timed out.")
14    except requests.exceptions.ConnectionError:
15        print("Error: Could not connect to API. Check your internet.")
16    except requests.exceptions.HTTPError:
17        print("Error: Invalid city or API key.")
18    except Exception as e:
19        print("Unexpected Error:", str(e))
20    return None
21 def fetch_weather(city):
22     data = get_weather_with_errors(city)
23     if data is None:
24         print("Error: City not found. Please enter a valid city.")
25         return None
26     result = {
27         "city": data["name"],
28         "temp": data["main"]["temp"],
29         "humidity": data["main"]["humidity"],
30         "weather": data["weather"][0]["description"].title()
31     }
32     return result
33 def save_weather(city):
34     result = fetch_weather(city)
35     if result is None:
36         return None
37     # Append to file
38     with open("results.txt", "a") as f:
39         f.write(json.dumps(result) + "\n")
40     print("Weather details saved to results.txt")
41     return result
42 before = len(open("results.txt").readlines())
43 save_weather("London")
44 after = len(open("results.txt").readlines())
45 assert after == before + 1
46 assert save_weather("xyz123") is None
47 assert "city" in save_weather("Delhi")
48
49

```